A FILTERED COLIMIT OF K-FLAT COMPLEXES IS K-FLAT

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Editorial note. The proof is the author's complete argument: the displayed identity supplies the reduction and exactness of filtered colimits supplies the conclusion.

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Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology and in chapter more-algebra. Those statements are the special cases of this ringed-site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- K-flat complex of O-modules. Same chapter, the definition of K-flatness by acyclicity of total tensor products with acyclic complexes. Supplies the property transferred from the members of the system to the colimit.
- Termwise filtered colimit of complexes of O-modules. Same chapter, the construction of the colimit of a directed system of complexes of O-modules termwise. Supplies the object colim_i K_i^bullet whose K-flatness is asserted.
- Exactness of filtered colimits. Same chapter, the fact that filtered colimits of abelian groups, and hence of complexes and of their cohomology, are exact. Supplies the step from the acyclicity of each member to the acyclicity of the colimit.

Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $\mathcal{K}_1^\bullet \to \mathcal{K}_2^\bullet \to \ldots$ be a system of K-flat complexes. Then $\operatorname{colim}_i \mathcal{K}_i^\bullet$ is K-flat.*

Proof. Because we are taking termwise colimits it is clear that

$$\operatorname{colim}_i \operatorname{Tot}(\mathcal{F}^\bullet \otimes_{\mathcal{O}} \mathcal{K}_i^\bullet) = \operatorname{Tot}(\mathcal{F}^\bullet \otimes_{\mathcal{O}} \operatorname{colim}_i \mathcal{K}_i^\bullet)$$

Hence the lemma follows from the fact that filtered colimits are exact. □

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